Junye Zhou

Portfolio (moonlitjaderabbit.github.io) | junye.zhou@mail.utoronto.ca

Education

University of Toronto Toronto, ON

Bachelor of Applied Science in Computer Engineering *Expected Graduation: 2030 (PEY Co-op)*

- Coursework: Algorithms & Data Structures (Python, C), Programming Fundamentals (C++), Digital Systems (Verilog), Electrical Circuits, Intro to Electronics.

Engineering Experience

Drone Team Member Sep 2026 – Present

RSX Robotics for Space Exploration

- Developed a **6 V** buck-converter design for the gripper servo and a **19 V** buck-boost converter design to power the Jetson Nano. Prepared the schematics in KiCad.
- Review battery voltage limits and the power needs of onboard components to guide regulator selection and power distribution planning.
- Performed soldering work with the electrical subteam. Review UART connections and SiK/MAVLink telemetry for the Kakute H7 flight controller to support ongoing communications work.

Team Lead Oct 2026

Trench–Hydro One Innovation Sprint | Winning Concept, **5th Place**

- Led a **4-person team** that proposed using electromagnetic and acoustic sensors to detect electrical discharges inside sealed capacitor voltage transformers.
- Used Python and openEMS, with support from AI coding tools, to simulate **300–700 MHz** electromagnetic signals through a simplified oil-filled model and examine the predicted fields at its outer housing.
- Evaluated detection thresholds in synthetic noise and interference tests; a higher threshold reduced false candidates from **8.93 to 1.02 per record**, clarifying the trade-off between false alarms and missed pulses.

Praxis Design Team Feb 2026 – Apr 2026

University of Toronto Praxis II

- Designed a swim-training concept that uses a rotating drum to control rope resistance, targeting **8 levels (60–170 N)** for the Mississauga Aquatic Club.
- Compared designs for adjustability, ease of manufacture, and poolside safety using Pugh matrices. Wrote requirements and test plans, then presented the team's concept at the Praxis II Showcase.

Bridge Design Team Nov 2025 – Jan 2026

University of Toronto CIV102 | Winning Design, **2nd Place**

- Designed, analyzed, and built a **1.25 m** matboard bridge with teammates, placing **2nd in the EngSci class**.
- Checked the team's Python structural analysis against hand calculations to assess the loads the bridge could carry and how it might fail.

Selected Project

Toronto Badminton Drop-in Finder Website University of Toronto

https://torontodropin.ca/

- Developed a website using AI coding tools and City of Toronto open data to help badminton players find nearby community centres and track drop-in sessions, serving **70 daily users**.

Technical Skills

Programming: C, C++, Python, Verilog, Java
Development: Git/GitHub, GCC (MSYS2), VS Code, ModelSim, DESim, Quartus Prime
Hardware & Analysis: KiCad, LTspice, TI WEBENCH, MATLAB, openEMS, soldering, circuit analysis

For project details and additional work, visit my portfolio: https://moonlitjaderabbit.github.io/